Probabilistic Prediction of Outcome and Herding Time in Single-Pursuer Multi-Evader Herding

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***Abstract*—A single pursuer can herd many non-cooperative evaders by circling a goal with a radius that follows one tracked evader. Existing theory for this law is local: it guarantees a stable herd near the equilibrium but says nothing about which starting configurations end in success, in a stalled herd, or in a lost evader, or how long herding takes. We learn this map from 60,000 validated simulations with classical, probabilistic machine learning. A Gaussian-process (GP) classifier on permutation-invariant features predicts the three-way outcome from the start alone with 93.7 % balanced accuracy and the best calibrated probabilities of all models, transfers to unseen flock sizes, and reveals that evaders are lost even where the stability condition of the theory holds. A kernel mixture-cure survival model handles runs that do not settle within the horizon and separates slow from stuck herds, which we confirm by re-simulating ten times longer. Finally, we propose a risk-aware active-learning rule that picks the next simulation by the Bayes risk of the decision; it reaches the decision cost of random sampling with 2.3× fewer simulations, whereas standard entropy sampling improves accuracy but worsens the decision cost.**

I. Introduction

Herding many evaders with one pursuer is a hard nonlinear, switched-like control problem. Singh and Chakraborty proposed a circular pursuer trajectory [1] and a feedback variant whose radius follows the evader that was farthest at the start [2]. Their theorems give existence, uniqueness and local asymptotic stability of a common equilibrium ring under a parameter condition, and an inner (Lyapunov-ellipse) estimate of the region of attraction for the tracked evader. For the other evaders, the region is only shown by brute-force simulation [2].

A practitioner needs more: *given this flock and these settings, will herding succeed, stall, or lose an evader, how sure are we, and how long will it take?* We treat this as a pattern-recognition problem with three contributions:

1) A validated dataset and calibrated probabilistic predictors (GP classification and regression, Bayes-risk decisions) of the outcome and herding time for any flock size, with an analysis of where the local theory fails.
2) A kernel mixture-cure survival model for herding time with right-censoring, which separates slow herds from herds that never settle.
3) Risk-aware active learning: choosing the next simulation by the Bayes risk of the downstream decision, and the finding that the usual uncertainty-based choice optimises the wrong target.

GP classification with active learning has estimated regions of attraction of single systems [3] and missile capture regions [4]; see also safe ROA learning [5]. We add three asymmetric-cost outcome classes, many agents with permutation-invariant inputs, censored times, and a decision-driven acquisition rule.

II. System and Data

A. Model

Evader i at $x_i \in \mathbb{R}^2$ moves away from the pursuer p with speed k/d_i^2, $d_i = \|x_i - p\|$; evaders do not interact [1]. The pursuer moves at angular speed ω with radius $R_p(t) = R\,e^{k_1(r_0(t)-\kappa)}$, where r_0 is the distance of the tracked evader (farthest at $t = 0$) and $\kappa = r_0(0)$ [2]; $k_1 = 0$ gives the fixed circle of [1] (our baseline law). In the frame rotating with the pursuer the dynamics are autonomous. Scaling lengths by $1/k_1$ and time by $1/\omega$ leaves two parameters, $q = k_1 R$ and $\beta = k k_1^3/\omega$, plus the scaled start; for the circle law only $\gamma = k/(\omega R^3)$ remains. The theory's condition is $a = 2q^2 e^{-2\kappa} > 1$ (in scaled units).

B. Labels

A run is *loss* if an evader leaves the pursuer circle at $T_{\max} = 300$ laps or goes beyond an escape radius; *success* if, from some lap t_s on, every evader stays within 1 % (radius) and 0.05 rad of the common equilibrium; otherwise *deadlock*. The herding time is $T_h = t_s$.

C. Simulation and data

We wrote our own simulator (RK4 with a step that shrinks during close passes). It reproduces the published ring radii [1], the equilibrium of the running example of [2], an independent loss count for the tracked-index law, and agrees with an adaptive DOP853 solver: 700 of 700 re-simulated runs get the same label and herding time. We generated 60,000 arXiv-law runs ($N = 1..6$, three strata of a: $a > 1$, $6/e^2 < a \le 1$, $a \le 6/e^2$; Latin-hypercube sampling [6]; plus boundary sets) and 10,000 circle-law runs: 26,454 success, 16,013 deadlock, 17,533 loss. A fixed stratified 20 % test set is used throughout.

D. Features

Only the start is used. Lengths are divided by κ. Global: N, $\log q$, $\log \beta$, κ, $\log a$, $\log(q/\kappa)$, $\log(r^*/\kappa)$. Flock: means, spreads and minima of radii, circular variance, largest angular gap, pairwise distances, hull area, angles relative to the pursuer

TABLE I
TEST-SET RESULTS (12,000 RUNS). BAL. ACC. = MEAN PER-CLASS RECALL.

model (training runs)	bal. acc.	log-loss	Brier	ECE
theory rule ($a > 1$)	0.457	–	–	–
logistic regression (48k)	0.825	0.482	0.258	0.039
SVM + Platt (4k)	0.918	0.254	0.130	0.034
SVM + Platt (20k)	0.938	0.200	0.099	0.021
GP classifier (4k)	0.937	**0.162**	**0.092**	0.028

and to the tracked evader, distances to the pursuer and to the equilibrium point. All are permutation-invariant and defined for every N (24 features).

III. METHODS

A. Classifiers and calibration

Baselines: majority class, the theory rule (success iff $a > 1$), logistic regression, and an RBF-SVM (with Platt scaling [7] for probabilities). The GP classifier uses one binary GP per class (Laplace approximation, one-vs-rest [8]) with an ARD kernel $\sigma_f^2 \exp(-\frac{1}{2}\sum_j (x_j - x'_j)^2/\ell_j^2)$. Exact GP cost is $O(n^3)$, so the kernel is tuned by marginal likelihood on 1,000 runs and the GP is fitted on 4,000. We cap $\sigma_f^2 \le 100$: without the cap, three of six random training sets produced an over-confident GP (balanced accuracy down to 0.885); with it, seven sets give 0.936–0.938.

B. Bayes-risk decisions

With cost matrix L (calling a true loss "success" costs 10; loss→deadlock and deadlock→success cost 3; other errors 1), the decision is $\hat{a}(x) = \arg\min_j \sum_i P(i \mid x)\, L_{ij}$.

C. Herding time with censoring

Let $y = \log(1 + T_h)$ and $c = \log(1 + T_{\max})$. With random Fourier features $z(x)$ [9] of the ARD kernel, $f(x) = b + w^\top z(x)$ approximates a GP. The Tobit model [10] uses $y \sim \mathcal{N}(f, \sigma^2)$ for settled runs and $P(y > c) = \Phi((f - c)/\sigma)$ for deadlocks. Our *kernel mixture-cure* model [11] adds $\pi(x) = \mathrm{sigmoid}(g(x))$, the probability that a run ever settles:

$$\mathcal{L} = \prod_{\text{settled}} \pi\, \mathcal{N}(y; f, \sigma^2) \prod_{\text{deadlock}} \left[(1 - \pi) + \pi\, \Phi\left(\tfrac{f - c}{\sigma}\right)\right], \quad (1)$$

with Gaussian priors on the feature weights (MAP by L-BFGS with analytic gradients).

D. Risk-aware active learning

Starting from 200 random runs, the GP repeatedly adds batches from a pool. *Entropy* sampling picks the largest predictive entropy [12]. *Risk-aware* sampling picks the largest Bayes risk of the best decision, $r(x) = \min_j \sum_i P(i \mid x) L_{ij}$, i.e. where acting is most expensive.

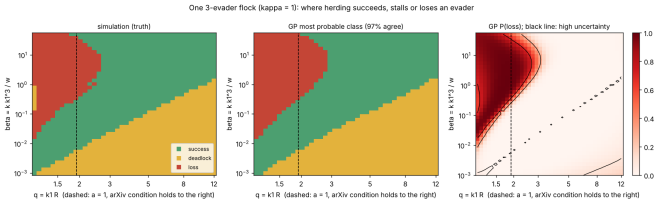


Fig. 1. One 3-evader flock: simulated outcome (left), GP prediction (middle) and GP $P(\text{loss})$ with its high-uncertainty contour (right) over q and β; dashed: $a = 1$.

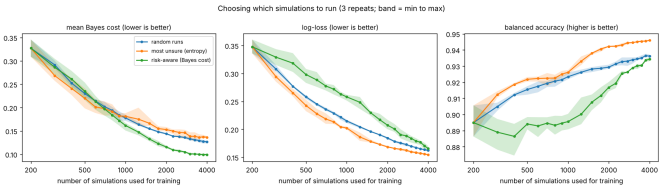


Fig. 2. Quality vs number of simulations for three acquisition rules (3 repeats; bands = min to max).

IV. RESULTS

A. Outcome prediction

Table I: the GP matches the SVM trained on five times more data in accuracy and gives the best probabilities. Trained on $N \le 4$ and tested on $N = 5, 6$, the SVM keeps 92.8 %. The theory rule reaches only 45.7 %: 52 % of successful runs violate $a > 1$ and 11 % of lost runs satisfy it. Uncertainty is informative: answering only the 80 % most confident runs raises GP accuracy to 98.6 %. The Bayes rule cuts "loss called success" from 169 to 9 test runs. Fig. 1 compares the learned map with 1,600 fresh simulations (97.3 % agreement): losses occur right of $a = 1$, where the theorems hold, because those theorems are local.

B. Herding time

On settled test runs a GP regression predicts T_h with median error 0.7 laps and 89 % coverage of its 90 % bands. Censoring: TBD.

C. Unsupervised regimes

PCA and a GMM (BIC) on start features do not recover the classes (ARI 0.08), but on trajectory shapes the clusters are 95 % pure and the first two components form one arm per class: the labels are natural regimes, and predicting them from the start requires supervision.

D. Risk-aware active learning

Fig. 2: risk-aware sampling reaches the decision cost of random sampling at 4,000 runs with 1,750 (2.3× fewer) and ends 22 % lower (0.099 vs 0.127). Entropy sampling reaches random's accuracy with 1,500 runs and ends best in accuracy (0.946), yet its decision cost is *worse* than random (0.137): what to simulate should follow the decision the model is for.

E. Theory comparison

TBD

V. Conclusion

TBD

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